

ML-KEM.Encaps(ek)

Uses the encapsulation key to generate a shared secret key and an associated ciphertext.

Checked input: encapsulation key $ek \in \mathbb{B}^{384k+32}$.

Output: shared secret key $K \in \mathbb{B}^{32}$.

Output: ciphertext $c \in \mathbb{B}^{32(d_u k + d_v)}$.

1: $m \xleftarrow{\$} \mathbb{B}^{32}$

▷ m is 32 random bytes

2: **if** $m == \text{NULL}$ **then**

3: **return** $\perp$

▷ return an error indication if random bit generation failed

4: **end if**

5: $(K, c) \leftarrow \text{ML-KEM.Encaps_internal}(ek, m)$

▷ run internal encapsulation algorithm

6: **return** (K, c)